

Supplementary Material for: Signatures of sexually antagonistic
intragenomic coevolution in *Drosophila melanogaster*

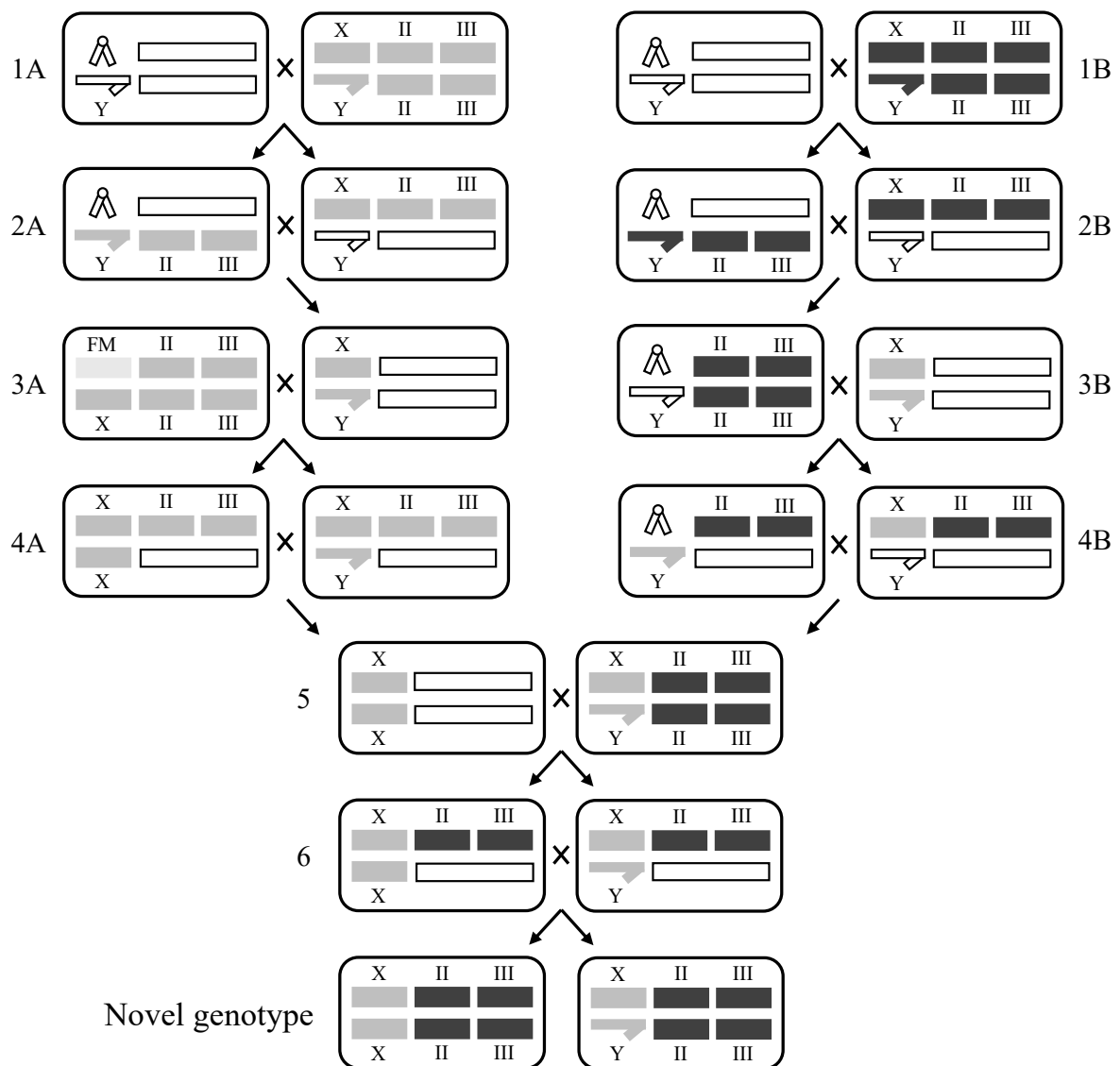
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S1 – Details of the crosses to generate the novel genotypes.



To generate the four novel genotypes, we used females from the clone-generator (CG) stock, with a double-X-chromosome (DX), a Y-chromosome, and two translocations of chromosome II and III (C(1)DX, *y, f*; T(2;3) *rdgC st in ri p^P bw^D*). All crosses were carried out at the same time, with Cwt and FLX/CFM starting either in the A or B lane of the crosses, depending on whether the regime was contributing sex chromosomes (A) or autosomes (B) to the final genotypes. As all crosses were done within each of the four replicate populations from the evolution experiment at the end of the crosses there were four new genotypes per replicate population. All females were collected as virgins.

For a walkthrough of the crosses, we will use two representative populations called grey and black. To produce e.g. FLXsex genotypes, FLX would be the grey population (A) and Cwt would be the black population (B). Autosomes (II and III) and chromosome X are depicted

as rectangles, the double-X-chromosome is depicted as $\mathbb{A}$, and the chromosome Y as a half arrow. The autosome translocation (2:3) is depicted as the elongated white rectangle.

Cross 1: CG females were crossed to either grey males (1A) or black males (1B). This cross was done with 200 breeding adults ($\text{♀}24:16\text{♂}$ per vial) in five vials and flipped once without reducing egg numbers. We did not reduce the number of eggs because half of the offspring will not reach adulthood because of uneven numbers of sex chromosomes.

Cross 2: The surviving offspring from cross 1A were crossed together in 2A and the same was done in 2B with the offspring from 1B. This cross was done with 224 breeding adults ($\text{♀}16:16\text{♂}$ per vial) in seven vials and flipped twice without reduction in egg numbers.

Cross 3: Males from 2A with grey sex chromosomes and two translocated balancers were crossed to females from the grey population (3A). For 3B, females with black autosomes and CG sex chromosomes were also crossed to males from 2A with grey sex chromosomes and two translocated balancers. Both crosses were done with 160 breeding adults ($\text{♀}16:16\text{♂}$ per vial) in five vials and flipped once without reduction in egg numbers.

Cross 4: Offspring from 3A were crossed to each other. These were males and females with two grey sex chromosomes and one set of grey autosomes and one translocated balancer (4A). The only surviving offspring from 3B were also crossed together. These were males or females with one grey sex chromosome and one set of grey autosomes and one translocated balancer (4B). Both crosses were done with 160 breeding adults ($\text{♀}16:16\text{♂}$ per vial) in five vials and flipped twice without reduction in egg numbers.

Cross 5: Females from 4A with two grey X chromosomes and two translocated balancers were crossed to males from 4B with two grey sex chromosomes and black autosomes. The cross was done with 160 breeding adults ($\text{♀}16:16\text{♂}$ per vial) in five vials and flipped once with a reduction in egg numbers to ~200.

Cross 6: The offspring from 5, both with grey sex chromosomes, one set of black chromosomes and one translocated balancer were crossed together. The cross was done with 160 breeding adults ($\text{♀}16:16\text{♂}$ per vial) in five vials and flipped twice with a reduction in egg numbers to ~200.

Novel genotypes: Offspring with grey sex chromosomes and black autosomes are used to establish novel genotype population with 160 breeding adults ($\text{♀}16:16\text{♂}$ per vial) in five vials.

S2 – Sperm competition assay

At day 9-11 from egg laying, target males were collected as virgins along with competitor males and females from the $\text{LH}_M\text{-}bw$ population [1]. The $\text{LH}_M\text{-}bw$ population is homozygous

for a recessive brown-eye marker, so that the offspring from a target red-eyed male mated to an LH_M-bw female will always be red-eyed, allowing us to measure paternity share. For sperm defence, one target male was paired in a test tube with one LH_M-bw female on day 14 for an hour to mate. The target male was then quickly removed under light CO_2 anaesthesia, and the female was left alone in the tube for 48 hours. The female was then introduced to a LH_M-bw male in a new test tube for 24 hours. The longer time interval for the second mating increases the chances of the female remating. After 24 hours the second male was quickly removed, and the female was left alone for an 18-hour window to lay eggs. The female was then discarded, and the tubes were left under standard conditions for 12 days, after which the offspring were counted and scored for eye-colour. We only collected data from females where we could confirm that the female had mated with the first male (by checking for larvae in the first test tube). For sperm offence the same procedure was followed, except that the first male was from the LH_M-bw population and the second male was the target male. We carried out both assays with 20 males per genotype per replicate population, in total 640 males assayed.

S3 – Statistical methods for the downstream analyses of differentially expressed transcripts.

All statistical analyses were conducted in R version 4.5.2 [2].

We wanted to test if there were a correlation between the number of significant transcripts in males and females. To do so we collected the number of transcripts in the 36 intersects that had significant transcripts in both sexes. We used a Shapiro-Wilk normality test (*shapiro.test*) to test for normality of the data, and both the male and female data was not normally distributed (Male: $P = 7.389 \times 10^{-08}$, female: $P = 1.242 \times 10^{-10}$). We therefore used the non-parametric Spearman's test to test for correlation (*cor.test*).

To test for sex biased expression, transcripts related to sexual conflict, sexually antagonistic transcripts, mitochondria-related transcripts, and chromosome location we used a Pearson's Chi-squared test (*chisq.test*) or a Fisher's exact test (*fisher.test*). We used available datasets to calculate the observed and expected numbers of the different categories in our dataset. All data sets used here have previously been shared with our group by the authors or downloaded as supplementary materials from the relevant papers.

To classify our transcripts as sex biased, we used the Sebida database [3]. Sebida characterizes a large number of genes as sex biased or unbiased in expression based on robust previous data. We used this information to determine both the observed numbers of

differentially expressed transcripts within each class, as well as the expected number based on the proportion of male-, female-, or unbiased transcripts across all transcripts present in our dataset. We then used Pearson's Chi-squared test to calculate if the observed numbers deviated from the random expectation. If the observed numbers were too small for a chi-squared approximation to be correct, we used Fisher's exact test instead. Because we tested both up- and down-regulated transcripts in seven groups we Bonferroni-adjusted the significance to $P > 0.004$ to account for multiple testing. This was the case for all downstream tests.

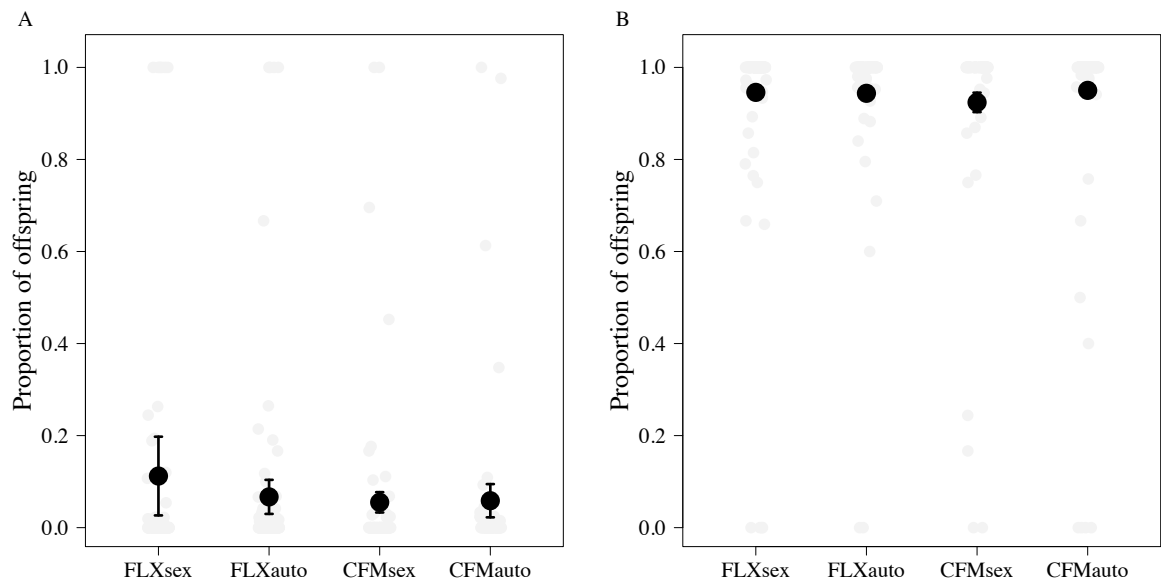
We used a similar approach for the tests for overrepresentation of genes previously associated with sexual conflict, sexual antagonism, and mitochondrial function. For sexual conflict, we used the datasets from two previous papers which classified genes in *D. melanogaster* [4] and *D. pseudoobscura* [5] as being related to sexual conflict. For sexual antagonism, we used two datasets [6,7] which have used our base population LH_M to identify transcripts whose expression has sexually antagonistic effects on fitness [6], as well as genetic loci that show signatures of sexual antagonism [7]. To classify our transcripts in one of three mitochondria-related categories we used three different data sets of mito-annotated [8], mito-proteome [9], and mito-sensitive [10] genes. As the *D. melanogaster* genome is well annotated information such as chromosome location is added automatically to the count table by the package *subread* [11].

To test for association of differentially expressed transcripts with sex specific fitness and sexual antagonism we did a two-tailed mean-rank gene set enrichment (*wilcoxGST* from the *limma* package [12]) as described by [13,14] and using the data from [6].

The data for the tissue specific expression analysis was downloaded from the FlyAtlas2 website [15]. Following the method in [6] we tested for overabundance of transcripts in each of the 18 adult tissues using an one-tailed Fisher's exact test and adjusted for multiple testing using a Bonferroni correction.

Gene Ontology (GO) terms were analysed using *enrichGO* from the package *clusterProfile* [16], which returns enriched GO categories after Bonferroni correction. We also used the REVIGO website [17] to help summarize common themes among the GO terms.

S4 – Sperm competition assays. A) Sperm defence. B) Sperm offence. The raw data is plotted as grey points and the black circles are the mean ($\pm$ SE) of fitted values from the models.



S5 – Results of statistical analysis of phenotypic data

A)

Sperm defence

Source	<i>df</i>	χ^2	<i>P</i>
Genotype	3	6.082	0.108
Block	1	57.229	3.879*10⁻¹⁴

B)

Sperm offence

Source	<i>df</i>	χ^2	<i>P</i>
Genotype	3	0.867	0.834
Block	1	0.270	0.604

C)

Locomotory activity

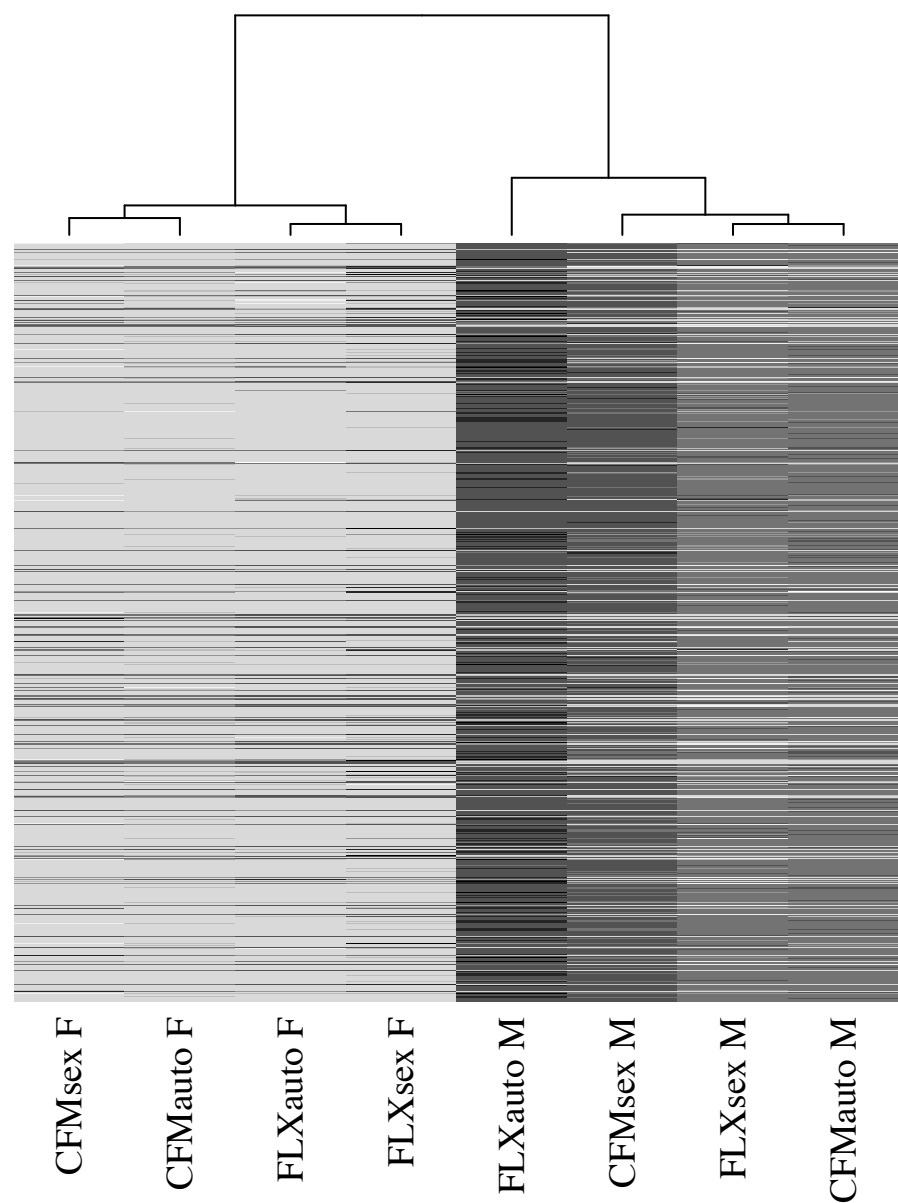
Source	<i>df</i>	χ^2	<i>P</i>
Genotype	3	4.643	0.200
Sex	1	5.832	0.016
Day	1	0.919	0.338
Genotype*Sex	3	22.024	6.448*10⁻⁰⁵

D)

Locomotory activity: post hoc tests

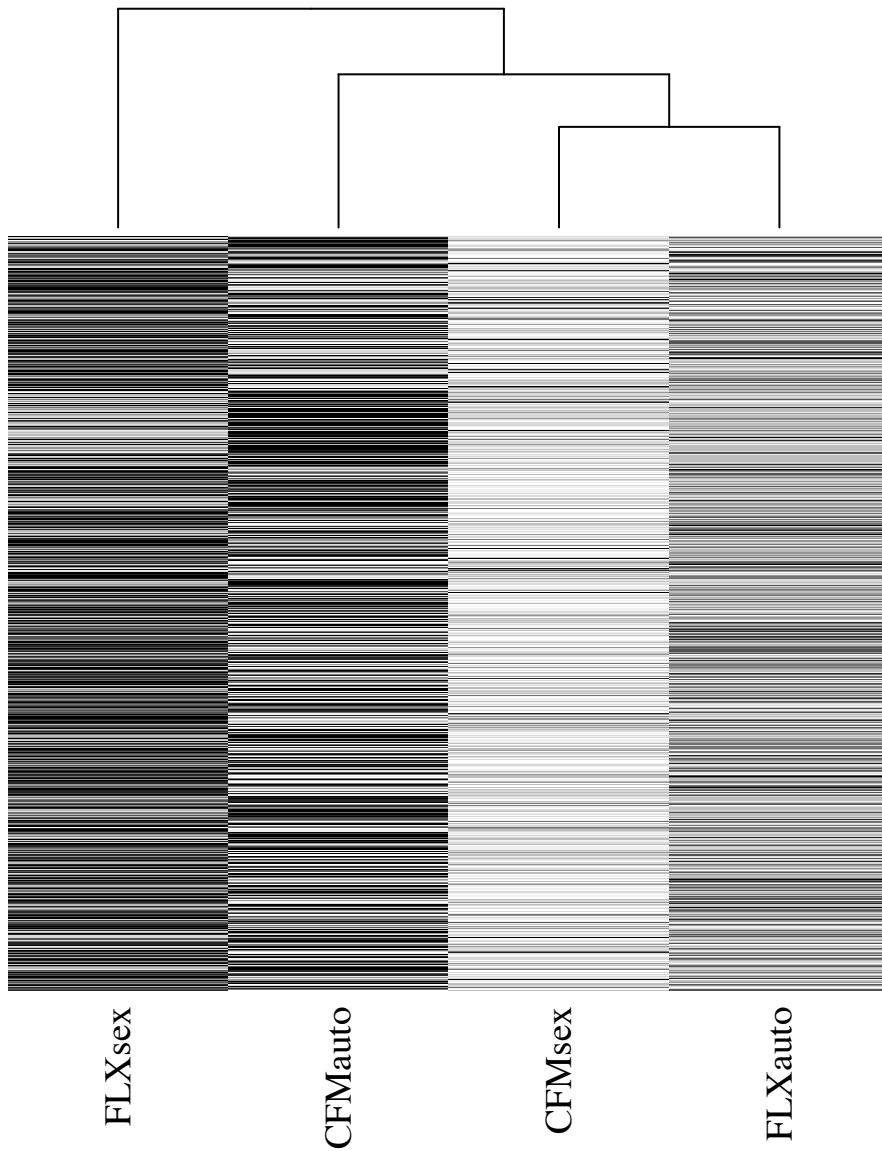
Source	<i>P</i>
Female	
FLXsex – FLXauto	0.992
FLXsex – CFMsex	0.811
FLXsex – CFMauto	0.037
FLXauto – CFMsex	0.651
FLXauto – CFMauto	0.018
CFMsex – CFMauto	0.323
Male	
FLXsex – FLXauto	0.990
FLXsex – CFMsex	0.893
FLXsex – CFMauto	0.779
FLXauto – CFMsex	0.731
FLXauto – CFMauto	0.911
CFMsex – CFMauto	0.364

S6 – Heat map with both sexes. There were 6961 significant differential expressed transcripts as an interaction between sex and genotype. The two sexes M and F are clearly separated with males having higher expression levels compared to females overall.

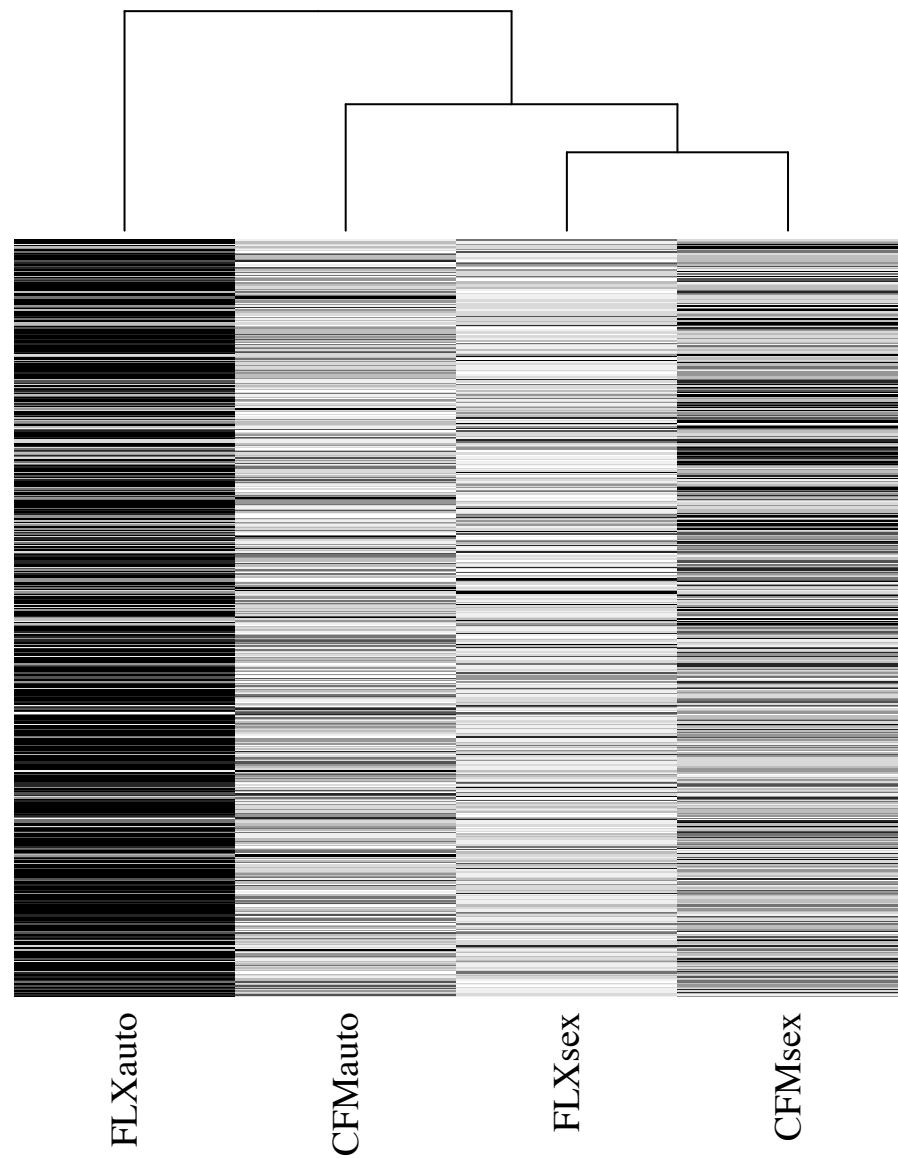


CFMsex F
CFMauto F
FLXauto F
FLXsex F
FLXauto M
CFMsex M
FLXsex M
CFMauto M

S7 – Heat map for females. There were 5725 significant transcripts in females between the four novel genotypes.



S8 – Heat map for males. There were 588 significant transcripts in males between the four novel genotypes.



S9 – Sex biased expression in the seven groups in females. The observed and expected numbers of up- or down-regulated transcripts as categorized as female-biased, male-biased, or un-biased using the Sebida database. P values in bold are significant deviation from random expectation.

Group	Sex bias	Up-regulated transcripts			Down-regulated transcripts		
		Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	Female	98	41.129	< 2.2*10⁻¹⁶	4	664.040	< 2.2*10⁻¹⁶
	Male	0	31.939		1220	515.667	
	Unbiased	12	36.933		552	596.293	
B	Female	469	193.305	< 2.2*10⁻¹⁶	10	117.404	< 2.2*10⁻¹⁶
	Male	0	150.113		98	91.171	
	Unbiased	48	173.583		206	105.426	
C	Female	146	59.450	< 2.2*10⁻¹⁶	9	112.917	< 2.2*10⁻¹⁶
	Male	0	46.166		137	87.687	
	Unbiased	13	53.384		156	101.397	
D	Female	170	67.675	< 2.2*10⁻¹⁶	2	88.613	< 2.2*10⁻¹⁶
	Male	1	52.554		112	68.814	
	Unbiased	10	60.771		123	79.573	
E	Female	53	32.529	8.298*10⁻⁰⁸	23	82.631	< 2.2*10⁻¹⁶
	Male	3	25.261		47	64.168	
	Unbiased	31	29.210		151	74.201	
F	Female	24	13.086	6.404*10⁻⁰⁵	2	86.744	< 2.2*10⁻¹⁶
	Male	0	10.162		118	67.362	
	Unbiased	11	11.751		112	77.894	
G	Female	67	30.286	< 2.2*10⁻¹⁶	9	60.945	< 2.2*10⁻¹⁶
	Male	0	23.519		61	47.328	

Unbiased	14	27.196	93	54.727
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S10 – Sex biased expression in the seven groups in males. The observed and expected numbers of up- or down-regulated transcripts as categorized as sex or un-biased using the Sebida database [3]. P values in bold are significant deviation from random expectation.

Group	Sex bias	Up-regulated transcripts			Down-regulated transcripts		
		Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	Female	0	0.374	0.626	6	5.608	0.765
	Male	0	0.290		3	4.355	
	Unbiased	1	0.336		6	5.036	
B	Female	7	12.712	0.007	12	19.817	1.524*10⁻⁰⁸
	Male	7	9.872		3	15.389	
	Unbiased	20	11.416		38	17.795	
C	Female	8	7.478	0.338	10	13.086	0.003
	Male	3	5.807		4	10.162	
	Unbiased	9	6.715		21	11.751	
D	Female	3	4.487	0.006	4	5.982	0.011
	Male	0	3.484		1	4.646	
	Unbiased	9	4.029		11	5.372	
E	Female	1	4.113	0.095	8	8.600	0.007
	Male	4	3.194		1	6.678	
	Unbiased	6	3.693		14	7.722	
F	Female	0	0.374	0.626	0	1.122	0.062
	Male	0	0.290		0	0.871	
	Unbiased	1	0.336		3	1.007	
G	Female	2	1.869	0.333	4	6.356	0.005

Male	0	1.452	1	4.936
Unbiased	3	1.679	12	5.708

S11 – Overall sex biased expression in sex-limited intersections. Test to see if there are more sex biased transcripts among the sex-limited significant groups than in the rest of the significant data set. P values in bold are significant.

Sex	Observed	Expected	<i>P</i> adjusted
Female	23	43.330	7.525*10⁻⁰⁷
Male	1	1.060	1.000

S12 – Sex biased expression in sex-limited intersections. The observed and expected numbers of up- or down-regulated transcripts as categorized as sex or un-biased using the Sebida database. P values in bold are significant deviation from random expectation.

Sex	Sex bias	Up-regulated transcripts			Down-regulated transcripts		
		Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
Female	Female	7	16.078	1.379*10⁻⁰⁸	1	10.469	0.001
	Male	3	12.485		12	8.130	
	Unbiased	33	14.437		15	9.401	
Male	Female	0	0.748	0.197	0	0.374	0.290
	Male	0	0.581		1	0.290	
	Unbiased	2	0.672		0	0.336	

S13 – Association with fitness in the seven groups in females. Up- or down-regulated transcripts as associated with male- or female-specific fitness related and sexually antagonistic transcripts characterized in [6]. P values in bold are significant.

Group	Up-regulated transcripts associated with			Down-regulated transcripts associated with		
	Female fitness	Male fitness	Sexual antagonism	Female fitness	Male fitness	Sexual antagonism
A	No association $P = 0.537$	No association $P = 0.442$	No association $P = 0.586$	No association $P = 0.847$	Positive $P = 0.002$	No association $P = 0.168$
B	Negative $P = 3.521 \times 10^{-09}$	Positive $P = 0.003$	Positive $P = 4.921 \times 10^{-07}$	Positive $P = 5.441 \times 10^{-05}$	No association $P = 0.029$	Negative $P = 3.186 \times 10^{-06}$
C	Negative $P = 4.295 \times 10^{-04}$	Positive $P = 7.961 \times 10^{-04}$	Positive $P = 3.062 \times 10^{-05}$	No association $P = 0.096$	No association $P = 0.780$	No association $P = 0.153$
D	Negative $P = 8.630 \times 10^{-04}$	No association $P = 0.237$	No association $P = 0.076$	Positive $P = 0.002$	No association $P = 0.022$	Negative $P = 1.993 \times 10^{-04}$
E	No association $P = 0.104$	No association $P = 0.195$	No association $P = 0.902$	Negative $P = 0.002$	No association $P = 0.587$	No association $P = 0.461$
F	No association $P = 0.344$	No association $P = 0.059$	No association $P = 0.824$	No association $P = 0.125$	No association $P = 0.876$	No association $P = 0.537$
G	No association $P = 0.242$	No association $P = 0.076$	No association $P = 0.171$	No association $P = 0.789$	No association $P = 0.944$	No association $P = 0.946$

S14 – Association with fitness in the seven groups in males. Up- or down-regulated transcripts as associated with male- or female-specific fitness related and sexually antagonistic transcripts characterized in [6]. P values in bold are significant.

Group	Up-regulated transcripts associated with			Down-regulated transcripts associated with		
	Female fitness	Male fitness	Sexual antagonism	Female fitness	Male fitness	Sexual antagonism
A	No association <i>P</i> = 0.193	No association <i>P</i> = 0.424	No association <i>P</i> = 0.431	No association <i>P</i> = 0.051	No association <i>P</i> = 0.021	No association <i>P</i> = 0.015
B	No association <i>P</i> = 0.060	No association <i>P</i> = 0.438	No association <i>P</i> = 0.223	No association <i>P</i> = 0.938	No association <i>P</i> = 0.300	No association <i>P</i> = 0.811
C	No association <i>P</i> = 0.145	No association <i>P</i> = 0.221	No association <i>P</i> = 0.752	No association <i>P</i> = 0.096	No association <i>P</i> = 0.623	No association <i>P</i> = 0.049
D	No association <i>P</i> = 0.158	No association <i>P</i> = 0.902	No association <i>P</i> = 0.137	Positive <i>P</i> = 1.679*10⁻⁰⁴	No association <i>P</i> = 0.032	Negative <i>P</i> = 3.058*10⁻⁰⁴
E	No association <i>P</i> = 0.745	No association <i>P</i> = 0.719	No association <i>P</i> = 0.565	No association <i>P</i> = 0.139	No association <i>P</i> = 0.175	No association <i>P</i> = 0.151
F	No association <i>P</i> = 0.728	No association <i>P</i> = 0.664	No association <i>P</i> = 0.550	No association <i>P</i> = 0.908	No association <i>P</i> = 0.546	No association <i>P</i> = 0.878
G	No association <i>P</i> = 0.787	No association <i>P</i> = 0.769	No association <i>P</i> = 0.538	No association <i>P</i> = 0.007	No association <i>P</i> = 0.627	No association <i>P</i> = 0.454

S15 –Association with change in the intensity of sexual conflict in the seven groups in females. The observed and expected numbers of up- or down-regulated transcripts as related to sexual conflict in A) *D. melanogaster* [4] and B) *D. pseudoobscura* [5]. P values in bold are significant deviation from random expectation.

A)

Group	Up-regulated transcripts			Down-regulated transcripts		
	Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	19	8.562	2.224*10⁻⁰⁴	178	134.131	8.850*10⁻⁰⁵
B	51	37.830	0.027	49	25.353	1.171*10⁻⁰⁶
C	12	11.747	0.939	42	24.689	3.114*10⁻⁰⁴
D	10	12.676	0.437	28	18.849	0.029
E	14	6.504	0.002	19	17.521	0.715
F	5	2.655	0.187	30	17.521	0.002
G	10	6.106	0.103	20	13.207	0.053

B)

Group	Up-regulated transcripts			Down-regulated transcripts		
	Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	16	13.895	0.550	292	217.693	9.734*10⁻⁰⁸
B	109	61.398	1.265*10⁻¹⁰	34	41.147	0.238
C	29	19.066	0.016	44	40.070	0.511
D	40	20.574	5.787*10⁻⁰⁶	39	30.591	0.108
E	7	10.556	0.247	35	28.437	0.193
F	5	4.309	0.614	33	28.437	0.365
G	8	9.910	0.521	21	21.435	0.921

S16 – Association with change in the intensity of sexual conflict in the seven groups in males. The observed and expected numbers of up- or down-regulated transcripts as related to sexual conflict in A) *D. melanogaster* [4] and B) *D. pseudoobscura* [5]. P values in bold are significant deviation from random expectation.

A)

Group	Up-regulated transcripts			Down-regulated transcripts		
	Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	0	0.265	1.000	4	1.062	0.018
B	3	3.584	1.000	11	3.849	0.001
C	6	1.792	0.007	12	2.456	2.697*10⁻⁰⁶
D	1	0.929	0.618	3	1.062	0.085
E	0	1.128	0.624	7	1.726	0.001
F	0	0.133	1.000	1	0.199	0.186
G	3	0.664	0.025	9	1.261	1.217*10⁻⁰⁶

B)

Group	Up-regulated transcripts			Down-regulated transcripts		
	Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	0	0.431	1.000	1	1.723	1.000
B	4	5.817	0.425	10	6.247	0.112
C	5	2.908	0.205	6	3.985	0.283
D	1	1.508	1.000	2	1.723	0.688
E	1	1.831	1.000	5	2.801	0.193
F	0	0.215	1.000	0	0.323	1.000
G	0	1.077	0.614	2	2.047	1.000

S17 – Sexual antagonistic transcripts in the seven groups in females. The observed and expected numbers of up- or down-regulated sexual antagonistic transcripts as described in A) [7] and B) [6]. P values in bold are significant deviation from random expectation.

A)

Group	Up-regulated transcripts			Down-regulated transcripts		
	Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	5	3.766	0.431	73	58.999	0.064
B	15	16.640	0.683	12	11.152	0.797
C	5	5.167	0.941	11	10.860	0.966
D	7	5.576	0.540	16	8.291	0.007
E	6	2.861	0.067	8	7.707	0.915
F	3	1.168	0.111	16	7.707	0.002
G	3	2.686	0.752	3	5.809	0.237

B)

Group	Up-regulated transcripts			Down-regulated transcripts		
	Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	15	10.398	0.137	233	162.908	1.021*10⁻⁰⁸
B	48	45.946	0.752	58	30.792	3.161*10⁻⁰⁷
C	21	14.268	0.063	41	29.986	0.036
D	14	15.396	0.711	37	22.893	0.002
E	15	7.900	0.008	18	21.280	0.458
F	3	3.224	1.000	30	21.280	0.049
G	9	7.416	0.544	25	16.041	0.020

S18 – Sexual antagonistic transcripts in the seven groups in males. The observed and expected numbers of up- or down-regulated sexual antagonistic transcripts as described in A) [7] and B) [6]. P values in bold are significant deviation from random expectation.

A)

Group	Up-regulated transcripts			Down-regulated transcripts		
	Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	0	0.117	1.000	1	0.467	0.378
B	0	1.576	0.410	4	1.693	0.089
C	1	0.788	0.551	3	1.080	0.093
D	1	0.409	0.340	0	0.467	1.000
E	0	0.496	1.000	0	0.759	1.000
F	0	0.058	1.000	0	0.088	1.000
G	1	0.292	0.257	1	0.555	0.431

B)

Group	Up-regulated transcripts			Down-regulated transcripts		
	Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	1	0.322	0.286	5	1.290	0.007
B	9	4.353	0.038	12	4.675	0.002
C	4	2.176	0.272	2	2.982	0.766
D	0	1.129	0.621	4	1.290	0.035
E	1	1.370	1.000	4	2.096	0.153
F	0	0.161	1.000	0	0.242	1.000
G	0	0.8069	1.000	3	1.532	0.194

S19 – Tissue-specific transcripts in female group A. P values in bold are significant.

Tissue	Up-regulated transcripts		Down-regulated transcripts	
	Odds ratio	<i>P</i> adjusted	Odds ratio	<i>P</i> adjusted
Accessory glands	0.232	> 1	0.932	> 1
Brain	0.703	> 1	4.152	8.443*10⁻¹¹⁰
Carcass	0.885	> 1	1.915	3.342*10⁻²⁰
Crop	2.276	0.009	1.553	9.998*10⁻⁰⁹
Eye	1.194	> 1	2.592	5.071*10⁻⁶¹
Fat body	1.175	> 1	1.111	> 1
Head	0.922	> 1	3.842	4.194*10⁻¹⁶²
Heart	1.323	> 1	1.567	1.205*10⁻⁰⁹
Hindgut	2.156	0.033	1.450	1.080*10⁻⁰⁵
Malpighian tubule	1.983	0.039	1.214	0.055
Mated spermatheca	1.570	> 1	1.008	> 1
Midgut	1.867	0.234	1.090	> 1
Ovary	2.924	8.136*10⁻⁰⁸	0.114	> 1
Rectal pad	1.911	0.189	1.538	2.056*10⁻⁰⁷
Salivary gland	2.297	0.025	1.077	> 1
Testes	0.131	> 1	1.354	1.939*10⁻⁰⁴
Thoracicoabdominal ganglion	0.969	> 1	3.585	6.407*10⁻¹²¹
Virgin spermatheca	2.088	> 1	0.547	> 1

S20 – Tissue-specific transcripts in male group A.

Tissue	Up-regulated transcripts		Down-regulated transcripts	
	Odds ratio	<i>P</i> adjusted	Odds ratio	<i>P</i> adjusted
Accessory glands	0.000	> 1	3.131	> 1
Brain	0.000	> 1	0.000	> 1
Carcass	0.000	> 1	0.960	> 1
Crop	0.000	> 1	0.000	> 1
Eye	0.000	> 1	0.436	> 1
Fat body	0.000	> 1	0.000	> 1
Head	0.000	> 1	0.342	> 1
Heart	0.000	> 1	2.139	> 1
Hindgut	5.544	> 1	0.000	> 1
Malpighian tubule	2.922	> 1	1.252	> 1
Mated spermatheca	0.000	> 1	0.000	> 1
Midgut	0.000	> 1	7.871	0.010
Ovary	0.000	> 1	0.000	> 1
Rectal pad	3.031	> 1	0.000	> 1
Salivary gland	0.000	> 1	2.077	> 1
Testes	0.000	> 1	0.763	> 1
Thoracicoabdominal ganglion	0.000	> 1	0.000	> 1
Virgin spermatheca	0.000	> 1	0.000	> 1

S21 – Tissue-specific transcripts in female group B. P values in bold are significant.

Tissue	Up-regulated transcripts		Down-regulated transcripts	
	Odds ratio	<i>P</i> adjusted	Odds ratio	<i>P</i> adjusted
Accessory glands	0.755	> 1	0.734	> 1
Brain	0.932	> 1	1.069	> 1
Carcass	1.105	> 1	2.072	1.020*10⁻⁰⁵
Crop	1.767	6.688*10⁻⁰⁵	2.104	5.325*10⁻⁰⁶
Eye	1.134	> 1	1.828	8.084*10⁻⁰⁵
Fat body	1.068	> 1	1.112	> 1
Head	0.853	> 1	1.660	2.024*10⁻⁰⁴
Heart	1.393	0.106	1.600	0.023
Hindgut	1.432	0.096	2.586	8.449*10⁻¹⁰
Malpighian tubule	1.342	0.175	2.169	1.299*10⁻⁰⁷
Mated spermatheca	1.118	> 1	1.252	> 1
Midgut	1.210	> 1	3.375	2.345*10⁻¹⁷
Ovary	6.062	1.260*10⁻⁹²	0.047	> 1
Rectal pad	1.511	0.031	2.199	3.850*10⁻⁰⁶
Salivary gland	1.485	0.073	1.977	4.619*10⁻⁰⁴
Testes	0.482	> 1	0.205	> 1
Thoracicoabdominal ganglion	1.239	0.702	1.765	2.667*10⁻⁰⁴
Virgin spermatheca	0.837	> 1	0.844	> 1

S22 – Tissue-specific transcripts in male group B. P values in bold are significant.

Tissue	Up-regulated transcripts		Down-regulated transcripts	
	Odds ratio	<i>P</i> adjusted	Odds ratio	<i>P</i> adjusted
Accessory glands	2.062	0.858	0.723	> 1
Brain	0.989	> 1	0	> 1
Carcass	0.294	> 1	3.928	0.004
Crop	1.305	> 1	2.144	> 1
Eye	0.893	> 1	1.785	> 1
Fat body	0.913	> 1	3.442	0.624
Head	0.570	> 1	0.855	> 1
Heart	0.386	> 1	4.390	6.493*10⁻⁰⁵
Hindgut	1.446	> 1	0.616	> 1
Malpighian tubule	1.195	> 1	2.650	0.068
Mated spermatheca	0.923	> 1	1.676	> 1
Midgut	0.547	> 1	7.871	1.718*10⁻⁰⁹
Ovary	0.176	> 1	0.239	> 1
Rectal pad	1.240	> 1	1.515	> 1
Salivary gland	1.000	> 1	1.500	> 1
Testes	0.645	> 1	0.176	> 1
Thoracicoabdominal ganglion	1.241	> 1	0.369	> 1
Virgin spermatheca	2.616	> 1	1.142	> 1

S23 – Tissue-specific transcripts in female group C. P values in bold are significant.

Tissue	Up-regulated transcripts		Down-regulated transcripts	
	Odds ratio	<i>P</i> adjusted	Odds ratio	<i>P</i> adjusted
Accessory glands	2.190	0.236	0.493	> 1
Brain	0.984	> 1	1.665	0.036
Carcass	1.295	> 1	2.722	1.240*10⁻¹¹
Crop	0.953	> 1	2.383	2.219*10⁻⁰⁸
Eye	1.627	0.164	2.265	3.653*10⁻⁰⁹
Fat body	0.859	> 1	2.561	0.001
Head	1.136	> 1	3.221	5.630*10⁻²⁵
Heart	2.102	0.004	2.157	1.038*10⁻⁰⁶
Hindgut	1.353	> 1	2.506	6.312*10⁻⁰⁹
Malpighian tubule	1.741	0.081	2.215	5.912*10⁻⁰⁸
Mated spermatheca	1.143	> 1	2.651	2.373*10⁻⁰⁴
Midgut	1.163	> 1	1.544	0.106
Ovary	3.846	9.907*10⁻¹⁷	0.009	> 1
Rectal pad	1.269	> 1	2.554	5.048*10⁻⁰⁹
Salivary gland	3.130	1.440*10⁻⁰⁶	1.358	> 1
Testes	0.451	> 1	0.138	> 1
Thoracicoabdominal ganglion	1.165	> 1	1.690	0.001
Virgin spermatheca	1.212	> 1	1.596	> 1

S24 Tissue-specific transcripts in male group C. P values in bold are significant.

Tissue	Up-regulated transcripts		Down-regulated transcripts	
	Odds ratio	<i>P</i> adjusted	Odds ratio	<i>P</i> adjusted
Accessory glands	0.751	> 1	0.829	> 1
Brain	2.584	> 1	0.000	> 1
Carcass	0.554	> 1	0.400	> 1
Crop	4.287	0.102	0.857	> 1
Eye	2.291	> 1	0.578	> 1
Fat body	0.000	> 1	0.000	> 1
Head	4.105	0.010	0.802	> 1
Heart	1.159	> 1	2.162	> 1
Hindgut	2.079	> 1	0.951	> 1
Malpighian tubule	3.690	0.082	2.817	0.196
Mated spermatheca	0.000	> 1	1.257	> 1
Midgut	1.640	> 1	19.239	3.533*10⁻¹⁵
Ovary	0.338	> 1	0.117	> 1
Rectal pad	2.066	> 1	0.520	> 1
Salivary gland	3.788	0.071	0.514	> 1
Testes	0.286	> 1	0.131	> 1
Thoracicoabdominal ganglion	5.212	0.001	0.000	> 1
Virgin spermatheca	0.000	> 1	0.000	> 1

S25 – Tissue-specific transcripts in female group D. P values in bold are significant.

Tissue	Up-regulated transcripts		Down-regulated transcripts	
	Odds ratio	<i>P</i> adjusted	Odds ratio	<i>P</i> adjusted
Accessory glands	2.005	0.420	0.430	> 1
Brain	0.610	> 1	1.976	0.004
Carcass	1.373	> 1	1.810	0.008
Crop	1.501	0.739	2.304	6.015*10⁻⁰⁶
Eye	1.736	0.040	2.118	6.695*10⁻⁰⁶
Fat body	1.614	> 1	1.076	> 1
Head	0.959	> 1	2.579	2.349*10⁻¹²
Heart	1.699	0.129	1.751	0.013
Hindgut	1.303	> 1	2.494	7.659*10⁻⁰⁷
Malpighian tubule	2.392	3.048*10⁻⁰⁵	2.580	3.774*10⁻⁰⁹
Mated spermatheca	0.870	> 1	1.188	> 1
Midgut	0.860	> 1	2.750	1.669*10⁻⁰⁸
Ovary	5.319	6.661*10⁻²⁸	0.037	> 1
Rectal pad	1.087	> 1	2.754	2.118*10⁻⁰⁸
Salivary gland	1.391	> 1	2.078	0.001
Testes	0.318	> 1	0.243	> 1
Thoracicoabdominal ganglion	0.880	> 1	1.808	0.001
Virgin spermatheca	0.551	> 1	0.938	> 1

S26 –Tissue-specific transcripts in male group D. P values in bold are significant.

Tissue	Up-regulated transcripts		Down-regulated transcripts	
	Odds ratio	<i>P</i> adjusted	Odds ratio	<i>P</i> adjusted
Accessory glands	2.817	> 1	0.626	> 1
Brain	3.411	> 1	0.758	> 1
Carcass	4.320	0.865	2.057	> 1
Crop	6.668	0.125	1.000	> 1
Eye	5.609	0.073	0.935	> 1
Fat body	3.729	> 1	0.000	> 1
Head	4.399	0.206	0.733	> 1
Heart	2.780	> 1	2.139	> 1
Hindgut	3.024	> 1	1.109	> 1
Malpighian tubule	2.629	> 1	2.023	> 1
Mated spermatheca	8.224	0.564	0.000	> 1
Midgut	0.000	> 1	28.852	1.323*10⁻⁰⁸
Ovary	0.000	> 1	0.000	> 1
Rectal pad	4.041	0.599	1.299	> 1
Salivary gland	2.699	> 1	0.600	> 1
Testes	1.017	> 1	0.528	> 1
Thoracicoabdominal ganglion	1.955	> 1	0.434	> 1
Virgin spermatheca	5.232	> 1	0.000	> 1

S27 – Tissue-specific transcripts in female group E. P values in bold are significant.

Tissue	Up-regulated transcripts		Down-regulated transcripts	
	Odds ratio	<i>P</i> adjusted	Odds ratio	<i>P</i> adjusted
Accessory glands	0.000	> 1	0.465	> 1
Brain	0.1263	> 1	1.721	0.087
Carcass	1.452	> 1	1.881	0.006
Crop	0.957	> 1	2.089	3.495*10⁻⁰⁴
Eye	1.464	> 1	2.038	6.813*10⁻⁰⁵
Fat body	2.394	0.855	1.165	> 1
Head	1.244	> 1	2.171	1.579*10⁻⁰⁷
Heart	2.673	0.002	2.075	2.509*10⁻⁰⁴
Hindgut	1.060	> 1	2.020	0.002
Malpighian tubule	1.248	> 1	1.568	0.105
Mated spermatheca	2.099	> 1	1.154	> 1
Midgut	4.768	6.454*10⁻⁰⁹	0.999	> 1
Ovary	0.728	> 1	0.726	> 1
Rectal pad	0.722	> 1	2.012	0.002
Salivary gland	0.963	> 1	1.633	0.204
Testes	0.000	> 1	1.209	> 1
Thoracicoabdominal ganglion	0.726	> 1	2.262	1.188*10⁻⁰⁶
Virgin spermatheca	3.377	0.206	1.223	> 1

S28 – Tissue-specific transcripts in male group E.

Tissue	Up-regulated transcripts		Down-regulated transcripts	
	Odds ratio	<i>P</i> adjusted	Odds ratio	<i>P</i> adjusted
Accessory glands	0.000	> 1	2.817	0.613
Brain	0.758	> 1	0.455	> 1
Carcass	0.960	> 1	0.576	> 1
Crop	1.000	> 1	1.250	> 1
Eye	0.436	> 1	0.262	> 1
Fat body	0.000	> 1	0.000	> 1
Head	0.733	> 1	0.000	> 1
Heart	0.618	> 1	1.685	> 1
Hindgut	1.109	> 1	1.386	> 1
Malpighian tubule	2.023	> 1	0.730	> 1
Mated spermatheca	0.000	> 1	0.000	> 1
Midgut	1.875	> 1	4.834	0.032
Ovary	0.281	> 1	0.169	> 1
Rectal pad	1.299	> 1	1.653	> 1
Salivary gland	0.600	> 1	1.636	> 1
Testes	1.040	> 1	0.298	> 1
Thoracicoabdominal ganglion	0.434	> 1	0.261	> 1
Virgin spermatheca	4.185	> 1	0.000	> 1

S29 – Tissue-specific transcripts in female group F. P values in bold are significant.

Tissue	Up-regulated transcripts		Down-regulated transcripts	
	Odds ratio	<i>P</i> adjusted	Odds ratio	<i>P</i> adjusted
Accessory glands	0.783	> 1	0.823	> 1
Brain	0.663	> 1	3.139	9.266*10⁻¹⁰
Carcass	1.396	> 1	3.66	2.518*10⁻¹⁵
Crop	1.392	> 1	3.287	1.285*10⁻¹²
Eye	0.542	> 1	3.273	3.094*10⁻¹⁵
Fat body	0.966	> 1	2.730	0.005
Head	0.564	> 1	4.910	5.255*10⁻³⁴
Heart	0.726	> 1	2.787	2.553*10⁻⁰⁹
Hindgut	2.293	0.891	1.850	0.012
Malpighian tubule	1.933	> 1	1.783	0.007
Mated spermatheca	0.847	> 1	2.975	3.725*10⁻⁰⁴
Midgut	3.727	0.022	1.099	> 1
Ovary	1.542	> 1	0.054	> 1
Rectal pad	1.627	> 1	2.900	8.682*10⁻⁰⁹
Salivary gland	3.756	0.033	1.757	0.065
Testes	0.000	> 1	0.438	> 1
Thoracicoabdominal ganglion	1.161	> 1	2.994	7.204*10⁻¹³
Virgin spermatheca	1.363	> 1	2.288	0.224

S30 – Tissue-specific transcripts in male group F.

Tissue	Up-regulated transcripts		Down-regulated transcripts	
	Odds ratio	<i>P</i> adjusted	Odds ratio	<i>P</i> adjusted
Accessory glands	0.000	> 1	18.778	0.467
Brain	0.000	> 1	0.000	> 1
Carcass	0.000	> 1	0.000	> 1
Crop	0.000	> 1	0.000	> 1
Eye	0.000	> 1	0.000	> 1
Fat body	0.000	> 1	0.000	> 1
Head	0.000	> 1	0.000	> 1
Heart	0.000	> 1	18.529	0.480
Hindgut	0.000	> 1	0.000	> 1
Malpighian tubule	0.000	> 1	0.000	> 1
Mated spermatheca	0.000	> 1	90.381	0.025
Midgut	0.000	> 1	26.227	0.259
Ovary	0.000	> 1	2.112	> 1
Rectal pad	0.000	> 1	0.000	> 1
Salivary gland	0.000	> 1	4.499	> 1
Testes	0.000	> 1	1.144	> 1
Thoracicoabdominal ganglion	0.000	> 1	0.000	> 1
Virgin spermatheca	0.000	> 1	124.839	0.013

S31 – Tissue-specific transcripts in female group G. P values in bold are significant.

Tissue	Up-regulated transcripts		Down-regulated transcripts	
	Odds ratio	<i>P</i> adjusted	Odds ratio	<i>P</i> adjusted
Accessory glands	0.000	> 1	0.153	> 1
Brain	1.329	> 1	0.871	> 1
Carcass	1.562	> 1	2.668	4.051*10⁻⁰⁶
Crop	2.146	0.111	1.578	0.387
Eye	1.475	> 1	1.829	0.012
Fat body	0.000	> 1	0.765	> 1
Head	1.134	> 1	2.037	7.990*10⁻⁰⁵
Heart	1.835	0.462	1.764	0.063
Hindgut	2.042	0.241	1.463	> 1
Malpighian tubule	1.232	> 1	1.743	0.049
Mated spermatheca	1.463	> 1	0.671	> 1
Midgut	2.105	0.191	1.877	0.045
Ovary	1.940	0.042	0.129	> 1
Rectal pad	1.821	0.733	1.618	0.390
Salivary gland	2.247	0.141	1.344	> 1
Testes	0.000	> 1	0.690	> 1
Thoracicoabdominal ganglion	1.344	> 1	1.536	0.261
Virgin spermatheca	1.746	> 1	0.266	> 1

S32 – Tissue-specific transcripts in male group G.

Tissue	Up-regulated transcripts		Down-regulated transcripts	
	Odds ratio	<i>P</i> adjusted	Odds ratio	<i>P</i> adjusted
Accessory glands	2.683	> 1	0.522	> 1
Brain	0.000	> 1	0.000	> 1
Carcass	0.000	> 1	0.800	> 1
Crop	1.876	> 1	0.834	> 1
Eye	0.000	> 1	0.770	> 1
Fat body	0.000	> 1	0.000	> 1
Head	0.642	> 1	0.285	> 1
Heart	1.159	> 1	2.471	> 1
Hindgut	4.752	> 1	0.000	> 1
Malpighian tubule	0.000	> 1	1.031	> 1
Mated spermatheca	0.000	> 1	5.322	> 1
Midgut	1.640	> 1	4.686	0.162
Ovary	0.528	> 1	0.235	> 1
Rectal pad	2.598	> 1	0.505	> 1
Salivary gland	0.000	> 1	1.687	> 1
Testes	0.286	> 1	0.610	> 1
Thoracicoabdominal ganglion	0.000	> 1	0.000	> 1
Virgin spermatheca	0.000	> 1	3.488	> 1

S33 – Gene Ontology categories enriched for down-regulated transcripts in female group A. Biological Process (BP), Cellular Component (CC), and Molecular Function (MF). Only significant GO terms are listed.

Ontology	ID	Description	Gene ratio	Background ratio	Q value	P adjusted
BP	GO:0007186	G protein-coupled receptor signalling pathway	96/1561	242/12687	2.222*10 ⁻²⁴	2.630*10 ⁻²⁴
BP	GO:0099537	Trans-synaptic signalling	113/1561	324/12687	1.219*10 ⁻²³	1.443*10 ⁻²³
BP	GO:0007600	Sensory perception	110/1561	392/12687	3.674*10 ⁻¹⁵	4.350*10 ⁻¹⁵
BP	GO:0050804	Modulation of chemical synaptic transmission	59/1561	159/12687	1.614*10 ⁻¹³	1.910*10 ⁻¹³
BP	GO:0009581	Detection of external stimulus	41/1561	89/12687	5.940*10 ⁻¹³	7.032*10 ⁻¹³
BP	GO:0009582	Detection of abiotic stimulus	41/1561	89/12687	5.940*10 ⁻¹³	7.032*10 ⁻¹³
BP	GO:0071482	Cellular response to light stimulus	29/1561	49/12687	1.320*10 ⁻¹²	1.562*10 ⁻¹²
BP	GO:0006811	Monoatomic ion transport	121/1561	488/12687	1.320*10 ⁻¹²	1.562*10 ⁻¹²
BP	GO:0009583	Detection of light stimulus	33/1561	65/12687	7.068*10 ⁻¹²	8.368*10 ⁻¹²
BP	GO:0007602	Phototransduction	31/1561	60/12687	2.088*10 ⁻¹¹	2.472*10 ⁻¹¹
BP	GO:0042391	Regulation of membrane potential	32/1561	66/12687	6.902*10 ⁻¹¹	8.171*10 ⁻¹¹
BP	GO:0098660	Inorganic ion transmembrane transport	79/1561	300/12687	1.702*10 ⁻⁰⁹	2.015*10 ⁻⁰⁹
BP	GO:0050808	Synapse organization	83/1561	324/12687	2.335*10 ⁻⁰⁹	2.765*10 ⁻⁰⁹
BP	GO:0007623	Circadian rhythm	51/1561	167/12687	2.124*10 ⁻⁰⁸	2.514*10 ⁻⁰⁸
BP	GO:0009416	Response to light stimulus	49/1561	161/12687	4.204*10 ⁻⁰⁸	4.977*10 ⁻⁰⁸
BP	GO:0051046	Regulation of secretion	33/1561	87/12687	5.056*10 ⁻⁰⁸	5.985*10 ⁻⁰⁸
BP	GO:0007626	Locomotory behaviour	55/1561	199/12687	1.806*10 ⁻⁰⁷	2.138*10 ⁻⁰⁷
BP	GO:0030534	Adult behaviour	48/1561	165/12687	2.745*10 ⁻⁰⁷	3.249*10 ⁻⁰⁷
BP	GO:0044057	Regulation of system process	27/1561	67/12687	3.014*10 ⁻⁰⁷	3.569*10 ⁻⁰⁷
BP	GO:0006836	Neurotransmitter transport	40/1561	130/12687	8.290*10 ⁻⁰⁷	9.814*10 ⁻⁰⁷
BP	GO:0007269	Neurotransmitter secretion	33/1561	99/12687	1.562*10 ⁻⁰⁶	1.849*10 ⁻⁰⁶
BP	GO:0099643	Signal release from synapse	33/1561	99/12687	1.562*10 ⁻⁰⁶	1.849*10 ⁻⁰⁶

BP	GO:0042592	Homeostatic process	99/1561	471/12687	1.562*10 ⁻⁰⁶	1.849*10 ⁻⁰⁶
BP	GO:0048499	Synaptic vesicle membrane organization	13/1561	20/12687	1.900*10 ⁻⁰⁶	2.249*10 ⁻⁰⁶
BP	GO:0051606	Detection of stimulus	64/1561	264/12687	1.964*10 ⁻⁰⁶	2.325*10 ⁻⁰⁶
BP	GO:0042752	Regulation of circadian rhythm	34/1561	106/12687	2.593*10 ⁻⁰⁶	3.069*10 ⁻⁰⁶
BP	GO:1903350	Response to dopamine	11/1561	15/12687	2.716*10 ⁻⁰⁶	3.216*10 ⁻⁰⁶
BP	GO:1903351	Cellular response to dopamine	11/1561	15/12687	2.716*10 ⁻⁰⁶	3.216*10 ⁻⁰⁶
BP	GO:0009187	Cyclic nucleotide metabolic process	16/1561	31/12687	4.031*10 ⁻⁰⁶	4.772*10 ⁻⁰⁶
BP	GO:0040011	Locomotion	60/1561	252/12687	7.638*10 ⁻⁰⁶	9.042*10 ⁻⁰⁶
BP	GO:0008037	Cell recognition	36/1561	123/12687	1.103*10 ⁻⁰⁵	1.306*10 ⁻⁰⁵
BP	GO:0099500	Vesicle fusion to plasma membrane	11/1561	17/12687	1.633*10 ⁻⁰⁵	1.933*10 ⁻⁰⁵
BP	GO:0030239	Myofibril assembly	20/1561	50/12687	1.864*10 ⁻⁰⁵	2.207*10 ⁻⁰⁵
BP	GO:0061564	Axon development	76/1561	355/12687	1.896*10 ⁻⁰⁵	2.244*10 ⁻⁰⁵
BP	GO:0008038	Neuron recognition	34/1561	117/12687	2.480*10 ⁻⁰⁵	2.936*10 ⁻⁰⁵
BP	GO:0043052	Thermotaxis	11/1561	18/12687	3.215*10 ⁻⁰⁵	3.807*10 ⁻⁰⁵
BP	GO:0055002	Striated muscle cell development	20/1561	52/12687	3.378*10 ⁻⁰⁵	3.999*10 ⁻⁰⁵
BP	GO:0046903	Secretion	55/1561	238/12687	4.992*10 ⁻⁰⁵	5.910*10 ⁻⁰⁵
BP	GO:0052652	Cyclic purine nucleotide metabolic process	12/1561	22/12687	5.199*10 ⁻⁰⁵	6.155*10 ⁻⁰⁵
BP	GO:0016545	Male courtship behaviour, veined wing vibration	10/1561	16/12687	6.717*10 ⁻⁰⁵	7.952*10 ⁻⁰⁵
BP	GO:0007155	Cell adhesion	51/1561	221/12687	1.135*10 ⁻⁰⁴	1.343*10 ⁻⁰⁴
BP	GO:0042052	Rhabdomere development	17/1561	45/12687	2.267*10 ⁻⁰⁴	2.684*10 ⁻⁰⁴
BP	GO:0060271	Cilium assembly	40/1561	163/12687	2.267*10 ⁻⁰⁴	2.684*10 ⁻⁰⁴
BP	GO:0030431	Sleep	27/1561	93/12687	2.378*10 ⁻⁰⁴	2.816*10 ⁻⁰⁴
BP	GO:0048858	Cell projection morphogenesis	87/1561	455/12687	2.745*10 ⁻⁰⁴	3.250*10 ⁻⁰⁴
BP	GO:0050795	Regulation of behaviour	31/1561	115/12687	2.833*10 ⁻⁰⁴	3.354*10 ⁻⁰⁴
BP	GO:0031338	Regulation of vesicle fusion	9/1561	15/12687	2.864*10 ⁻⁰⁴	3.391*10 ⁻⁰⁴
BP	GO:0099504	Synaptic vesicle cycle	30/1561	111/12687	3.476*10 ⁻⁰⁴	4.115*10 ⁻⁰⁴

BP	GO:0044782	Cilium organization	42/1561	181/12687	0.001	0.001
BP	GO:0007622	Rhythmic behaviour	31/1561	122/12687	0.001	0.001
BP	GO:0042749	Regulation of circadian sleep/wake cycle	17/1561	50/12687	0.001	0.001
BP	GO:0055074	Calcium ion homeostasis	19/1561	60/12687	0.001	0.001
BP	GO:0003012	Muscle system process	14/1561	37/12687	0.001	0.001
BP	GO:0060322	Head development	36/1561	152/12687	0.001	0.001
BP	GO:0006874	Intracellular calcium ion homeostasis	18/1561	56/12687	0.001	0.001
BP	GO:0050890	Cognition	39/1561	174/12687	0.002	0.002
BP	GO:0007522	Visceral muscle development	8/1561	15/12687	0.002	0.002
BP	GO:0009612	Response to mechanical stimulus	17/1561	54/12687	0.002	0.002
BP	GO:0042761	Very long-chain fatty acid biosynthetic process	10/1561	23/12687	0.002	0.003
BP	GO:0007498	Mesoderm development	25/1561	97/12687	0.003	0.003
CC	GO:0043005	Neuron projection	139/1524	470/12923	7.586×10^{-24}	1.025×10^{-23}
CC	GO:0045202	Synapse	103/1524	329/12923	1.373×10^{-19}	1.855×10^{-19}
CC	GO:0098797	Plasma membrane protein complex	57/1524	165/12923	6.770×10^{-13}	9.149×10^{-13}
CC	GO:0097060	Synaptic membrane	39/1524	89/12923	1.254×10^{-12}	1.694×10^{-12}
CC	GO:0034702	Monoatomic ion channel complex	42/1524	118/12923	3.826×10^{-10}	5.170×10^{-10}
CC	GO:0016028	Rhabdomere	23/1524	43/12923	9.387×10^{-10}	1.269×10^{-09}
CC	GO:0005929	Cilium	58/1524	201/12923	9.387×10^{-10}	1.269×10^{-09}
CC	GO:0099503	Secretory vesicle	35/1524	95/12923	4.889×10^{-09}	6.607×10^{-09}
CC	GO:0030017	Sarcomere	28/1524	66/12923	6.396×10^{-09}	8.643×10^{-09}
CC	GO:0099081	Supramolecular polymer	54/1524	217/12923	7.456×10^{-07}	1.008×10^{-06}
CC	GO:0043679	Axon terminus	28/1524	85/12923	2.452×10^{-06}	3.313×10^{-06}
CC	GO:0036477	Somatodendritic compartment	59/1524	282/12923	5.657×10^{-05}	7.645×10^{-05}
CC	GO:0008076	Voltage-gated potassium channel complex	8/1524	12/12923	9.107×10^{-05}	1.231×10^{-04}
CC	GO:1990351	Transporter complex	56/1524	269/12923	1.000×10^{-04}	1.352×10^{-04}

CC	GO:0043235	Receptor complex	23/1524	77/12923	1.163*10 ⁻⁰⁴	1.571*10 ⁻⁰⁴
CC	GO:0016027	inaD signalling complex	7/1524	10/12923	1.893*10 ⁻⁰⁴	2.558*10 ⁻⁰⁴
CC	GO:0016528	Sarcoplasm	7/1524	13/12923	0.002	0.002
CC	GO:0044297	Cell body	36/1524	171/12923	0.002	0.003
CC	GO:0016529	Sarcoplasmic reticulum	6/1524	10/12923	0.002	0.003
CC	GO:0098831	Presynaptic active zone cytoplasmic component	6/1524	10/12923	0.002	0.003
CC	GO:0019897	Extrinsic component of plasma membrane	14/1524	45/12923	0.002	0.003
MF	GO:0038023	Signalling receptor activity	115/1525	433/12510	2.214*10 ⁻¹⁴	2.871*10 ⁻¹⁴
MF	GO:0005261	Monoatomic cation channel activity	60/1525	185/12510	2.151*10 ⁻¹¹	2.789*10 ⁻¹¹
MF	GO:0015318	Inorganic molecular entity transmembrane transporter activity	92/1525	371/12510	5.259*10 ⁻¹⁰	6.819*10 ⁻¹⁰
MF	GO:0015267	Channel activity	93/1525	426/12510	2.224*10 ⁻⁰⁷	2.883*10 ⁻⁰⁷
MF	GO:0005509	Calcium ion binding	52/1525	198/12510	8.696*10 ⁻⁰⁷	1.127*10 ⁻⁰⁶
MF	GO:0141108	Transporter regulator activity	16/1525	34/12510	1.016*10 ⁻⁰⁵	1.318*10 ⁻⁰⁵
MF	GO:0016247	Channel regulator activity	15/1525	33/12510	3.580*10 ⁻⁰⁵	4.641*10 ⁻⁰⁵
MF	GO:0004497	Monooxygenase activity	32/1525	112/12510	3.731*10 ⁻⁰⁵	4.837*10 ⁻⁰⁵
MF	GO:0005544	Calcium-dependent phospholipid binding	9/1525	13/12510	3.983*10 ⁻⁰⁵	5.164*10 ⁻⁰⁵
MF	GO:0005506	Iron ion binding	40/1525	156/12510	4.135*10 ⁻⁰⁵	5.360*10 ⁻⁰⁵
MF	GO:0016705	Oxidoreductase activity, acting on paired donors, with incorporation or reduction of molecular oxygen	42/1525	175/12510	1.312*10 ⁻⁰⁴	1.701*10 ⁻⁰⁴
MF	GO:0050839	Cell adhesion molecule binding	23/1525	76/12510	2.590*10 ⁻⁰⁴	3.358*10 ⁻⁰⁴
MF	GO:0015108	Chloride transmembrane transporter activity	15/1525	39/12510	3.007*10 ⁻⁰⁴	3.898*10 ⁻⁰⁴
MF	GO:0009975	Cyclase activity	12/1525	27/12510	3.289*10 ⁻⁰⁴	4.264*10 ⁻⁰⁴
MF	GO:0004016	Adenylate cyclase activity	8/1525	13/12510	3.656*10 ⁻⁰⁴	4.739*10 ⁻⁰⁴
MF	GO:0046906	Tetrapyrrole binding	33/1525	132/12510	3.721*10 ⁻⁰⁴	4.824*10 ⁻⁰⁴

MF	GO:0000981	DNA-binding transcription factor activity, RNA polymerase II-specific	88/1525	481/12510	4.871*10 ⁻⁰⁴	0.001
MF	GO:0106389	Ecdysteroid 22-kinase activity	17/1525	51/12510	0.001	0.001
MF	GO:0042802	Identical protein binding	32/1525	133/12510	0.001	0.001
MF	GO:0009922	Fatty acid elongase activity	9/1525	20/12510	0.002	0.003

S34 – Gene Ontology categories enriched for transcripts in female group B. Biological Process (BP), Cellular Component (CC), and Molecular Function (MF). Only significant GO terms are listed.

Ontology	ID	Description	Up-regulated transcripts		Background ratio	Q value	P adjusted
			Gene ratio				
BP	GO:0043161	Proteasome-mediated ubiquitin-dependent protein catabolic process	37/468		242/12687	2.530*10 ⁻¹⁰	2.77*10 ⁻¹⁰
BP	GO:0043632	Modification-dependent macromolecule catabolic process	42/468		331/12687	1.165*10 ⁻⁰⁹	1.28*10 ⁻⁰⁹
BP	GO:0022613	Ribonucleoprotein complex biogenesis	34/468		332/12687	1.546*10 ⁻⁰⁵	1.69*10 ⁻⁰⁵
BP	GO:0016072	rRNA metabolic process	23/468		178/12687	3.212*10 ⁻⁰⁵	3.52*10 ⁻⁰⁵
BP	GO:0032543	Mitochondrial translation	18/468		133/12687	2.987*10 ⁻⁰⁴	3.27*10 ⁻⁰⁴
BP	GO:0140053	Mitochondrial gene expression	20/468		165/12687	4.278*10 ⁻⁰⁴	4.69*10 ⁻⁰⁴
BP	GO:0006364	rRNA processing	19/468		157/12687	0.001	0.001
BP	GO:0000398	mRNA splicing, via spliceosome	27/468		294/12687	0.001	0.001
BP	GO:0045184	Establishment of protein localization	37/468		490/12687	0.003	0.003
BP	GO:0009894	Regulation of catabolic process	28/468		329/12687	0.003	0.003
CC	GO:0000502	Proteasome complex	21/480		56/12923	6.552*10 ⁻¹⁴	8.437*10 ⁻¹⁴
CC	GO:0005681	Spliceosomal complex	22/480		194/12923	1.209*10 ⁻⁰⁴	1.557*10 ⁻⁰⁴
CC	GO:0005768	Endosome	27/480		274/12923	1.272*10 ⁻⁰⁴	1.637*10 ⁻⁰⁴
CC	GO:0099023	Vesicle tethering complex	10/480		45/12923	1.349*10 ⁻⁰⁴	1.737*10 ⁻⁰⁴

CC	GO:0044391	Ribosomal subunit	20/480	176/12923	2.295*10 ⁻⁰⁴	2.956*10 ⁻⁰⁴
CC	GO:0098798	Mitochondrial protein-containing complex	19/480	172/12923	0.001	0.001
CC	GO:0000313	Organellar ribosome	12/480	84/12923	0.001	0.001
CC	GO:0005761	Mitochondrial ribosome	12/480	84/12923	0.001	0.001
CC	GO:0030008	TRAPP complex	5/480	13/12923	0.001	0.001
CC	GO:0120114	Sm-like protein family complex	13/480	103/12923	0.002	0.002
CC	GO:0030684	Preribosome	11/480	78/12923	0.002	0.002
MF	GO:0043024	Ribosomal small subunit binding	5/435	10/12510	0.003	0.003
MF	GO:0140098	Catalytic activity, acting on RNA	24/435	264/12510	0.003	0.003
MF	GO:0030234	Enzyme regulator activity	35/435	481/12510	0.003	0.003
MF	GO:0008135	Translation factor activity, RNA binding	11/435	73/12510	0.003	0.003
MF	GO:0180051	Translation factor activity	11/435	73/12510	0.003	0.003

Down-regulated transcripts

Ontology	ID	Description	Gene ratio	Background ratio	Q value	P adjusted
BP	GO:0019646	Aerobic electron transport chain	17/277	99/12687	3.993*10 ⁻⁰⁸	4.259*10 ⁻⁰⁸
BP	GO:0042775	Mitochondrial ATP synthesis coupled electron transport	17/277	105/12687	3.993*10 ⁻⁰⁸	4.259*10 ⁻⁰⁸
CC	GO:0098803	Respiratory chain complex	16/269	107/12923	7.899*10 ⁻⁰⁸	8.505*10 ⁻⁰⁸
CC	GO:1990204	Oxidoreductase complex	17/269	125/12923	7.899*10 ⁻⁰⁸	8.505*10 ⁻⁰⁸
CC	GO:0005743	Mitochondrial inner membrane	23/269	300/12923	4.644*10 ⁻⁰⁶	5.000*10 ⁻⁰⁶
MF	GO:0009055	Electron transfer activity	9/269	40/12510	4.212*10 ⁻⁰⁵	4.775*10 ⁻⁰⁵

S35 – Gene Ontology categories enriched for transcripts in male group B. Biological Process (BP) and Molecular Function (MF). Only significant GO terms are listed.

Up-regulated transcripts						
Ontology	ID	Description	Gene ratio	Background ratio	Q value	<i>P</i> adjusted
BP	GO:0098869	Cellular oxidant detoxification	3/31	13/12687	0.001	0.001
BP	GO:0045454	Cell redox homeostasis	4/31	20/12687	0.001	0.002
MF	GO:0051920	Peroxiredoxin activity	7/31	13/12510	4.079*10 ⁻⁰⁵	8.913*10 ⁻⁰⁵
MF	GO:0016209	Antioxidant activity	3/31	55/12510	7.605*10 ⁻⁰⁵	1.662*10 ⁻⁰⁴
MF	GO:0008233	Peptidase activity	2/31	453/12510	0.001	0.001

Down-regulated transcripts						
Ontology	ID	Description	Gene ratio	Background ratio	Q value	<i>P</i> adjusted
MF	GO:0005506	Iron ion binding	7/44	156/12510	8.978*10 ⁻⁰⁵	1.162*10 ⁻⁰⁴

S36 – Gene Ontology categories enriched for transcripts in female group C. Cellular Component (CC), and Molecular Function (MF). Only significant GO terms are listed.

Up-regulated transcripts						
Ontology	ID	Description	Gene ratio	Background ratio	Q value	<i>P</i> adjusted
CC	GO:0042175	Nuclear outer membrane-endoplasmic reticulum membrane network	20/144	323/12923	8.195*10 ⁻⁰⁸	9.766*10 ⁻⁰⁸
CC	GO:0005789	Endoplasmic reticulum membrane	19/144	318/12923	2.099*10 ⁻⁰⁷	2.502*10 ⁻⁰⁷
CC	GO:0098588	Bounding membrane of organelle	15/144	431/12923	0.003	0.003
MF	GO:0008135	Translation factor activity, RNA binding	7/136	73/12510	0.002	0.002

MF	GO:0180051	Translation factor activity	7/136	73/12510	0.002	0.002
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Down-regulated transcripts

Ontology	ID	Description	Gene ratio	Background ratio	Q value	P adjusted
MF	GO:0001653	Peptide receptor activity	11/268	67/12510	4.125×10^{-05}	4.705×10^{-05}
MF	GO:0008188	Neuropeptide receptor activity	9/268	45/12510	4.125×10^{-05}	4.705×10^{-05}

S37 – Gene Ontology categories enriched for up-regulated transcripts in female group D. Biological Process (BP), Cellular Component (CC), and Molecular Function (MF). Only significant GO terms are listed.

Ontology	ID	Description	Gene ratio	Background ratio	Q value	P adjusted
BP	GO:0032543	Mitochondrial translation	17/168	133/12687	2.153×10^{-09}	2.242×10^{-09}
BP	GO:0140053	Mitochondrial gene expression	17/168	165/12687	3.621×10^{-08}	3.771×10^{-08}
CC	GO:0044391	Ribosomal subunit	23/174	176/12923	2.442×10^{-14}	2.981×10^{-14}
CC	GO:0005761	Mitochondrial ribosome	15/174	84/12923	1.805×10^{-11}	2.203×10^{-11}
CC	GO:0098798	Mitochondrial protein-containing complex	19/174	172/12923	6.790×10^{-11}	8.288×10^{-11}
MF	GO:0003735	Structural constituent of ribosome	23/157	290/12510	3.683×10^{-10}	4.131×10^{-10}

S38 – Gene Ontology categories enriched for down-regulated transcripts in male group D. Molecular Function (MF). Only significant GO terms are listed.

Ontology	ID	Description	Gene ratio	Background ratio	Q value	P adjusted
MF	GO:0008233	Peptidase activity	6/16	453/12510	3.374×10^{-04}	0.001

S39 – Gene Ontology categories enriched for up-regulated transcripts in female group E. Biological Process (BP). Only significant GO terms are listed.

Ontology	ID	Description	Gene ratio	Background ratio	Q value	<i>P</i> adjusted
BP	GO:0090066	Regulation of anatomical structure size	9/67	234/12687	0.003	0.003

S40 – Gene Ontology categories enriched for transcripts in female group F. Biological Process (BP), Cellular Component (CC), and Molecular Function (MF). Only significant GO terms are listed.

Up-regulated transcripts						
Ontology	ID	Description	Gene ratio	Background ratio	Q value	<i>P</i> adjusted
CC	GO:0033181	Plasma membrane proton-transporting V-type ATPase complex	3/31	17/12923	4.110*10 ⁻⁰⁴	0.001
CC	GO:0016471	Vacuolar proton-transporting V-type ATPase complex	3/31	29/12923	0.001	0.001
CC	GO:0090533	Cation-transporting ATPase complex	3/31	42/12923	0.001	0.002
CC	GO:0098533	ATPase dependent transmembrane transport complex	3/31	48/12923	0.001	0.002
MF	GO:0044769	ATPase activity, coupled to transmembrane movement of ions, rotational mechanism	3/31	32/12510	0.002	0.003
MF	GO:0046961	Proton-transporting ATPase activity, rotational mechanism	3/31	32/12510	0.002	0.003

Down-regulated transcripts						
Ontology	ID	Description	Gene ratio	Background ratio	Q value	<i>P</i> adjusted
BP	GO:0007186	G protein-coupled receptor signalling pathway	17/199	242/12687	3.104*10 ⁻⁰⁴	3.428*10 ⁻⁰⁴
BP	GO:0007268	Chemical synaptic transmission	18/199	319/12687	0.001	0.001

BP	GO:0060047	Heart contraction	6/199	33/12687	0.002	0.002
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S41 – Gene Ontology categories enriched for down-regulated transcripts in male group F. Biological Process (BP) and Molecular Function (MF). Only significant GO terms are listed.

Ontology	ID	Description	Gene ratio	Background ratio	Q value	<i>P</i> adjusted
BP	GO:0030968	Endoplasmic reticulum unfolded protein response	2/3	22/12687	1.418*10 ⁻⁰⁴	0.001
BP	GO:0006986	Response to unfolded protein	2/3	37/12687	1.418*10 ⁻⁰⁴	0.001
BP	GO:0036503	ERAD pathway	2/3	46/12687	1.418*10 ⁻⁰⁴	0.001
BP	GO:0034976	Response to endoplasmic reticulum stress	2/3	96/12687	0.001	0.002
MF	GO:0140662	ATP-dependent protein folding chaperone	2/3	31/12510	1.874*10 ⁻⁰⁵	3.382*10 ⁻⁰⁴
MF	GO:0044183	Protein folding chaperone	2/3	48/12510	2.271e*10 ⁰⁵	4.099*10 ⁻⁰⁴

S42 – Gene Ontology categories enriched for transcripts in the female group G. Biological Process (BP) and Cellular Component (CC). Only significant GO terms are listed.

Up-regulated transcripts

Ontology	ID	Description	Gene ratio	Background ratio	Q value	<i>P</i> adjusted
CC	GO:0042600	Egg chorion	6/65	42/12923	6.313*10 ⁻⁰⁶	8.152*10 ⁻⁰⁶

Down-regulated transcripts

Ontology	ID	Description	Gene ratio	Background ratio	Q value	<i>P</i> adjusted
BP	GO:0006091	Generation of precursor metabolites and energy	15/129	260/12687	6.362*10 ⁻⁰⁵	6.701*10 ⁻⁰⁵

S43 – Transcripts related to the mitochondria in the seven groups in females. The observed and expected numbers of up- or down-regulated transcripts as related to three classes of mitochondria transcripts. A) Mito-annotated transcripts [8], B) Mito-proteome transcripts [9], and C) Mito-sensitive transcripts [10]. P values in bold are significant deviation from random expectation.

A)

Group	Up-regulated transcripts			Down-regulated transcripts		
	Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	6	3.597	0.178	41	56.350	0.038
B	24	15.893	0.039	21	10.651	0.001
C	10	4.935	0.033	3	10.372	0.0152
D	18	5.325	2.540*10⁻⁰⁸	4	7.918	0.158
E	1	2.732	0.530	14	7.361	0.013
F	2	1.115	0.307	5	7.361	0.378
G	4	2.565	0.328	13	5.549	0.001

B)

Group	Up-regulated transcripts			Down-regulated transcripts		
	Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	7	5.895	0.641	91	92.351	0.886
B	62	26.046	5.535*10⁻¹³	32	17.456	3.659*10⁻⁰⁴
C	24	8.088	1.020*10⁻⁰⁸	15	16.999	0.620
D	33	8.728	< 2.2*10⁻¹⁶	11	12.978	0.574
E	5	4.478	0.806	18	12.064	0.080
F	4	1.828	0.108	10	12.064	0.543
G	7	4.204	0.200	22	9.093	1.180*10⁻⁰⁵

C)

Group	Up-regulated transcripts			Down-regulated transcripts		
	Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	6	9.384	0.251	217	147.015	2.046*10⁻⁰⁹
B	55	41.464	0.029	20	27.788	0.125
C	16	12.876	0.366	14	27.061	0.009
D	15	13.894	0.758	9	20.659	0.008
E	4	7.129	0.224	16	19.204	0.448
F	3	2.910	0.766	21	19.204	0.670
G	4	6.692	0.280	18	14.476	0.336

S44 – Transcripts related to the mitochondria in the seven groups in males. The observed and expected numbers of up- or down-regulated transcripts as related to three classes of mitochondria transcripts. A) Mito-annotated transcripts [8], B) Mito-proteome transcripts [9], and C) Mito-sensitive transcripts [10]. P values in bold are significant deviation from random expectation.

A)

Up-regulated transcripts				Down-regulated transcripts		
Group	Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	0	0.112	1.000	1	0.446	0.364
B	1	1.506	1.000	4	1.617	0.078
C	1	0.753	0.534	0	1.032	0.627
D	0	0.390	1.000	0	0.446	1.000
E	0	0.474	1.000	0	0.725	1.000
F	0	0.056	1.000	0	0.084	1.000
G	0	0.279	1.000	2	0.530	0.097

B)

Up-regulated transcripts				Down-regulated transcripts		
Group	Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	0	0.183	1.000	1	0.731	0.527
B	0	2.468	0.179	9	2.650	0.001
C	0	1.234	0.634	1	1.691	1.000
D	0	0.640	1.000	0	0.731	1.000
E	1	0.777	0.549	0	1.188	0.631
F	0	0.091	1.000	2	0.137	0.006
G	0	0.457	1.000	0	0.868	1.000

C)

Up-regulated transcripts				Down-regulated transcripts		
Group	Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	0	0.291	1.000	2	1.164	0.326
B	4	3.928	1.000	3	4.219	0.799
C	2	1.964	1.000	3	2.692	0.749
D	1	1.018	1.000	2	1.164	0.326
E	0	1.237	0.631	1	1.891	1.000
F	0	0.145	1.000	2	0.218	0.015
G	1	0.727	0.530	1	1.382	1.000

S45 – Chromosomal location of transcripts of in the seven groups in females. The observed and expected numbers of the chromosomal location of the up- or down-regulated transcripts. P values in bold are significant deviation from random expectation.

Group	Chromosome	Up-regulated transcripts			Down-regulated transcripts		
		Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	X	35	19.813	0.002	272	310.409	0.014
	2	43	51.283		813	803.439	
	3	49	57.042		914	893.656	
	4	2	0.861		22	13.496	
B	X	76	87.547	0.239	57	58.672	0.992
	2	241	226.601		154	151.862	
	3	247	252.046		169	168.915	
	4	6	3.806		2	2.551	
C	X	35	27.032	0.309	39	56.982	0.017
	2	70	69.968		151	147.489	
	3	71	77.825		176	164.051	
	4	0	1.175		5	2.478	
D	X	10	29.336	6.555*10⁻⁰⁵	37	43.620	0.687
	2	76	75.931		113	112.903	
	3	105	84.457		133	125.581	
	4	0	1.275		1	1.897	
E	X	18	15.052	0.273	40	40.395	0.429
	2	31	38.959		116	104.554	
	3	48	43.334		105	116.295	
	4	1	0.654		2	1.756	
F	X	8	6.144	0.783	41	40.395	0.704

	2	15	15.902		102	104.554	
	3	17	17.687		117	116.295	
	4	0	0.267		3	1.756	
G	X	16	14.130	0.214	32	30.565	0.215
	2	32	36.574		67	79.112	
	3	42	40.681		100	87.995	
	4	2	0.614		0	1.329	

S46 – Chromosomal location of transcripts of in the seven groups in males. The observed and expected numbers of the chromosomal location of the up- or down-regulated transcripts.

Group	Chromosome	Up-regulated transcripts			Down-regulated transcripts		
		Observed	Expected	<i>P</i> adjusted	Observed	Expected	<i>P</i> adjusted
A	X	1	0.614	0.815	3	2.457	0.775
	2	1	1.590		5	6.361	
	3	2	1.769		8	7.075	
	4	0	0.027		0	0.107	
B	X	5	8.294	0.067	7	8.908	0.879
	2	24	21.467		24	23.058	
	3	23	23.878		27	25.647	
	4	2	0.361		0	0.387	
C	X	2	4.147	0.659	7	5.683	0.872
	2	12	10.734		14	14.709	
	3	13	11.939		16	16.361	
	4	0	0.180		0	0.247	
D	X	1	2.150	0.123	2	2.457	0.950

	2	5	5.566		6	6.361	
	3	7	6.191		8	7.075	
	4	1	0.093		0	0.107	
E	X	0	2.457		4	3.993	
	2	12	6.361	0.034	12	10.336	0.812
	3	4	7.075		10	11.497	
	4	0	0.107		0	0.174	
F	X	0	0.307		1	0.461	
	2	1	0.795	1.000	1	1.193	0.557
	3	1	0.884		1	1.327	
	4	0	0.013		0	0.020	
G	X	1	1.536		3	2.918	
	2	5	3.975	0.918	9	7.553	0.808
	3	4	4.422		7	8.402	
	4	0	0.067		0	0.127	

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